MILITARY SPECIFICATION

COUPLING ASSEMBLY, HOSE, CHEMICAL, BIOLOGICAL, RADIOLOGICAL PROTECTIVE

This specification is approved for use within the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of a coupling to be used with and form part of the Protective Assembly, Helicopter Aircrewman, Chemical, Biological, Radiological (PASS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Box, Folding, Paperboard.
PPP-B-636	-	Box, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 53), Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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FEDERAL

PPP-C-795	-	Cushioning Material, Flexible, Cellular, Plastic Film for Packaging Applications.
PPP-C-1842	-	Cushioning Material, Plastic, Open Cell (For Packaging Applications).

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-P-117	-	Bags, Sleeves and Tubing - Interior Packaging.
MIL-A-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys.
MIL-N-18307	-	Nomenclature & Identification for Aeronautical Systems Including Joint Electronics Type Designated Systems & Associated Support Systems
MIL-V-85830	-	Ventilator, Tactical, Helicopter Aircrewman, Chemical, Biological, Radiological Protective.
MIL-M-85831	-	Mask, Helicopter Aircrewman, Chemical, Biological, Radiological Protective, Non-Oxygen.
MIL-P-85833	-	Protective Assembly, Helicopter Aircrewman, Chemical, Biological, Radiological.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

DOD-STD-100	-	Engineering Drawing Practices.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-143	-	Order of Precedence for the Selection of Standards and Specifications.

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MIL-STD-210	-	Climatic Extremes for Military Equipment.
MIL-STD-282	-	Filter Units, Protection Clothing Gas Mask Components and Related Products, Performance Test Methods.
DOD-STD-480	-	Configuration Control - Engineering Changes, Deviations and Waivers.
MIL-STD-483	-	Configuration Management Practices for Systems, Equipment, Munitions and Computer Programs.
MIL-STD-490	-	Specification Practices.
MIL-STD-794	-	Procedures for Packaging and Packing of Parts and Equipment.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1189	-	Standard Department of Defense Bar Code Symbology.

HANDBOOKS

MIL-HDBK-695	-	Rubber Products: Recommended Shelf Life.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Naval Air Systems Command

SD-24	-	General Specification for Design and Construction of Aircraft Weapons Systems.

Headquarters, Department of the Army

Army Field Manual FM 3-5	-	NBC Decontamination

(Copies of specifications, standards, handbooks, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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AEROSPACE MATERIALS SPECIFICATIONS

AMS-3238 - Butyl Rubber, Phosphate Ester Resistant.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 3951 - Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Couplings furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for the opening of bids (see 4.3 and 6.3).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected as specified in MIL-STD-143.

3.4 Materials. Materials referenced herein shall conform to applicable specifications and standards. Materials that are not covered by specifications and that are not specifically described herein shall be the best quality and the lightest practical weight. Materials shall be corrosion resistant, or suitably treated to resist corrosion due to fuels, salt spray, environmental, chemical or biological warfare conditions.

3.4.1 Aluminum alloy. Aluminum alloy parts shall be anodically treated as specified in MIL-A-8625.

3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are as specified in MIL-STD-889.

3.4.3 Nonmetallic materials. Rubber seals and o-rings shall meet the requirements as specified in AMS-3238 and the requirements specified herein.

3.4.4 Fungus-proof materials. All materials shall be incapable of supporting fungoid growth.

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3.4.5 Age. Elastomeric components shall be comprised of materials with expected shelf storage life of greater than 5 years in accordance with MIL-HDBK-695 when consistent with the properties required for functional suitability. These components alone or as part of an assembly shall be delivered to the Government with less than 18 months elapsed from date of cure/manufacture to the date of delivery to the Government.

3.4.6 Toxic materials. The use of materials which produce toxic effects under any conditions shall be avoided.

3.4.7 Flammable materials. The use of materials which will support combustion shall be avoided.

3.5 Design and Construction. The coupling shall be used with and form part of the protective assembly (PASS), as shown on Figure 1. The coupling shall meet the requirements as specified in MIL-P-85833 and this specification. The coupling shall be designed and constructed as shown on Figure 2, for installation in line between the mask, MIL-M-85831, and ventilator, MIL-V-85830. The coupling shall have a quick disconnect fitting, for emergency egress quick disconnection. The coupling shall have a locking mechanism to avoid inadvertant disconnection. The design and construction shall permit decontamination, using procedures of Army FM 3-5, without damage to or deterioration of the coupling or its protective capability. The coupling shall be compatable with oxygen and oxygen systems of naval aviation. The coupling is intended for use in a contaminated environment and shall be designed to prevent the ingress of all forms of contamination.

3.5.1 Storage life. The coupling shall be designed to meet the performance requirements of this specification after five years storage under worldwide climatic conditions as specified in MIL-STD-210 when packaged as specified in MIL-STD-794.

3.5.2 Interchangeability. All parts, subassemblies and assemblies having the same part number shall be identical as specified in DOD-STD-480 and DOD-STD-100. When any unit or part is interchanged, the coupling shall meet all performance requirements without adjustment or modification of any part or subassembly.

3.6 Hardware.

3.6.1 Fasteners. All fasteners, including bolts, screws, nuts, rivets and washers shall be as specified in SD-24.

3.7 Performance.

3.7.1 Separation force.

3.7.1.1 Unlocked. When tested as specified in 4.8.2.1.1, the axial force required to separate the coupling at the quick disconnect fitting shall be not greater than five pounds.

3.7.1.2 Locked. When tested as specified in 4.8.2.1.2, the coupling shall not separate at the quick disconnect fitting while in the locked position under a minimum axial force of 100 pounds.

3.7.2 Self sealing valve. When tested as specified in 4.8.2.2, the force required to open the self-sealing valve, contained within the male portion of the coupling, shall be not greater than four pounds.

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3.7.3 Air leakage. The overall leakage rate of the coupling shall not exceed .006 $in.^3$/min. when tested for vacuum or pressure, as specified in 4.8.2.3.1 and 4.8.2.3.2.

3.7.4 Pressure drop. The pressure drop through the coupling shall be not greater than 0.5 in. water gage (WG.) when tested as specified in 4.8.2.4.

3.7.5 Water leakage. The self-sealing valve shall be water-tight (zero leakage) when tested as specified in 4.8.2.5.

3.7.6 Noise level. The noise level, when tested as specified in 4.8.2.6, for both components of the test, shall be determined by 1/3 octave band analysis over the frequency range 200-5000 Hz. The noise level with the coupling in line shall be equal to or less than that noise level measured without the coupling in line for all frequencies.

3.7.7 Chemical agent permeation. The coupling seals and o-ring materials shall have an endpoint time of not less than 360 minutes when tested as specified in 4.8.2.7.1, 4.8.2.7.2 and 4.8.2.7.3.

3.7.8 Endurance. Prior to and upon completion of the endurance test specified in 4.8.3, the coupling shall meet the requirements of 3.7.1, 3.7.2, 3.7.3, 3.7.4 and 3.7.5. without failure after 500 cycles, when tested as specified in 4.8.3.

3.7.9 Environmental tests. Prior to and at the completion of each environmental test, the coupling shall be tested and shall meet the requirements of 3.7.1, 3.7.2, 3.7.3, 3.7.4 and 3.7.5.

3.7.9.1 High temperature. When tested as specified in 4.8.4.1, the coupling shall operate without failure.

3.7.9.2 Low temperature. When tested as specified in 4.8.4.2, the coupling shall operate without failure.

3.7.9.3 Salt fog. When tested as specified in 4.8.4.3, the coupling shall withstand both the corrosive and physical effects of the salt fog environment and operate without failure.

3.7.9.4 Icing and freezing rain. When tested as specified as in 4.8.4.4, the coupling shall withstand the effects of icing produced by a freezing rain and shall operate without failure.

3.7.9.5 Fungus. When tested as specified in 4.8.4.5, the coupling shall not support fungal growth and shall operate without failure.

3.7.9.6 Low pressure and high humidity. When tested as specified in 4.8.4.6, the coupling shall operate without failure.

3.7.9.7 Dust. When tested as specified in 4.8.4.7, the coupling shall withstand the effect of blowing dust and shall operate without failure.

3.7.9.8 Normal acceleration. When tested as specified in 4.8.4.8, the coupling shall not inadvertantly disconnect and shall operate without failure.

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3.7.9.9 High temperature, diurnal cycle and high humidity. When tested as specified in 4.8.4.9, the coupling shall operate without failure.

3.8 Components. The coupling shall consist of the components specified in 3.8.1 and 3.8.2 and other parts as necessary which are essential to form a complete, assembled, and operating coupling.

3.8.1 Male portion. The male portion (see figure 2) of the coupling shall be attached to the air inlet hose of the mask and shall contain a self sealing valve which automatically closes when the two portions are separated. It shall be possible to open the self-sealing valve with a chemical, biological, radiological (CBR) protective gloved hand to allow air to flow freely into the mask.

3.8.2 Female portion. The female portion (see figure 2) of the coupling shall be attached to the ventilator delivery hose and shall contain a single action, one hand operated locking mechanism to prevent inadvertent disconnection and allow for quick disconnection with one Chemical, biological, radiological (CBR) gloved hand.

3.9 Dimensions. The overall length of the coupling shall be not greater than five in. The outside diameter shall be not greater than 1.5 in.

3.10 Weight. The weight of the coupling shall be not greater than eight ounces.

3.11 Color. The coupling shall be colored black, approximately matching color number 37038 as specified in FED-STD-595.

3.12 Nameplates and identification markings. Nameplate approval, equipment identification marking and serial number assignment shall be as specified in MIL-N-18307.

3.13 Configuration. The coupling furnished shall be produced under a program for managing system configuration as specified in DOD-STD-480, MIL-STD-483 and MIL-STD-490.

3.14 Workmanship. The coupling shall be free from contamination and imperfections such as dirt, grease, oil, foreign matter, malfunctioning of hardware and damage such as cracks, burrs, sharp edges, rough grinding marks, and uneven surfaces due to forming operations. Particular attention shall be given to accuracy of dimensions; radii of fillets; marking of parts and assemblies; thoroughness of welding, brazing, painting, swaging, and soldering; alignment of parts; and tightness of assembly screws and bolts. Rubber parts shall be free from damage, cuts and tears.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. First article inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Quality conformance verification inspection (see 4.6).

4.3 Qualification inspection. Qualification inspection shall be performed in the order specified in Table I, consisting of all the tests and examinations listed in this specification.

4.3.1 Qualification samples. Qualification samples shall consist of 15 couplings. Samples shall be forwarded to a test facility set forth in the letter of authorization to submit samples (see 6.3). The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Samples submitted by (name) (date) for qualification inspection as specified in MIL-C-85829 and number under authorization (reference authorizing letter and number) (see 6.3).

4.3.1.1 Disposition of samples. Upon completion of qualification inspection, all samples shall be consumed or destroyed and shall not be considered as part of the quantity to be delivered under contract.

4.3.2 Retention of qualification. To retain qualification, the contractor shall forward a report at six month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of a summary of the results of the tests performed, indicating the number of samples that have passed and the number that have failed during the six month period. If the summary of the test results indicates non conformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list. Failure to submit the report within 30 days after the end of each six month period may result in loss of qualification for the product. In addition to the periodic reports, the contractor shall immediately inform the qualifying activity of inspection data indicating failure of the qualified product to meet this specification. In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods, there has been no production, the manufacturer may be required at the discretion of the qualifying activity, to submit the qualified products to testing in accordance with qualification inspection requirement and the reason for no production.

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4.4 First article inspection. First article inspection shall be performed in the order specified in Table II.

4.4.1 First article samples. After award of the contract or order, the manufacturer shall submit three couplings. The samples shall be representative of the construction, workmanship and materials to be used during production. When a manufacturer is in continuous production of these couplings from contract to contract, submission of further first article samples on a new contract may be waived at the discretion of the acquiring activity (see 6.2.1). Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2.1).

4.4.1.1 Disposition of samples. Upon completion of the first article inspection, one approved coupling shall be returned to the manufacturer for use in monitoring production. The other samples shall be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of all the examinations and tests specified in Table III. The sampling and inspection levels shall be as specified in MIL-STD-105.

4.5.1 Inspection lot. An inspection lot size shall be expressed in units of one coupling made by one manufacturer under essentially the same conditions and from the same materials and components.

4.5.1.1 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing couplings, fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing couplings, fully prepared for delivery, with the exception that the shipping container need not be sealed.

4.5.2 Sampling for tests and examination of couplings. The sample size, acceptance criteria, tests and examinations required for the coupling shall be as specified in Table III.

4.6 Quality conformance verification inspection. At the option of the Government and upon completion of the quality conformance inspection requirements of 4.5, a random sample, one for every 150 or fraction thereof, shall be selected from each lot of couplings (see 6.2.1). Each coupling, selected as a sample unit, shall be forwarded to a laboratory designated at the time of the award. The designated laboratory shall conduct any or all the tests and examinations of 4.8 of this specification. The serial numbers of the units in the lot, represented by the sample units, shall be included with the data accompanying the samples to the laboratory. The Government activity responsible for conducting the quality conformance verification program shall report the results of the tests and examinations to the designated inspection and acceptance activity specified in the acquisition document. Final acceptance of the lot from which the sample units were selected shall be based upon successful completion of the inspection program by the cognizant Quality Assurance Representative/ Specialist at the contractor's facility; applying the applicable acceptance criteria specified in Table III.

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4.6.1 Rejected lots. If an inspection lot is rejected, the manufacturer may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection and shall not thereafter be tendered for acceptance unless the former rejection or requirement of correction is disclosed. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.7 Test conditions.

4.7.1 Temperature and pressure. Unless otherwise specified in the contract, inspections shall be conducted at local ambient temperature and barometric pressure. Corrections shall be made to provide agreement with the temperature and pressure calibration of the instruments. Inspection data provided by an instrument not calibrated to normal temperature and pressure (NTP) conditions shall be corrected to determine NTP conditions. NTP conditions are 70°F and 29.92 in. Hg.

4.8 Inspection methods.

4.8.1 Visual examination.

4.8.1.1 Coupling. Every coupling selected as a sample unit from the lot shall be visually examined for major and minor defects to determine conformance to this specification. The classification of defects, (see Table IV), shall be used to classify the defects found.

4.8.1.1.1 Dimensions. Each coupling selected as a sample unit from the lot shall be checked dimensionally to determine conformance to the dimensions specified herein.

4.8.1.2 Packaging. Each of the fully prepared shipping containers, containing couplings, selected as a sample unit from the lot, shall be examined to determine that the packaging, packing and marking conform to this specification. The classification of defects, (see Table V), shall be used to classify the defects found.

4.8.2 Performance.

4.8.2.1 Separation force.

4.8.2.1.1 Unlocked. A separation force shall be applied to the coupling with the locking mechanism moved to the unlocked position. The axial force required to separate the coupling at the quick disconnect fitting shall be as specified in 3.7.1.1.

4.8.2.1.2 Locked. An axial force up to 100 pounds shall be applied to the coupling with the locking mechanism moved to the locked position. The coupling shall not separate at a lower force as specified in 3.7.1.2.

4.8.2.2 Self sealing valve. A opening force shall be applied to the self sealing valve. The force required to open the valve shall be as specified in 3.7.2.

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4.8.2.3 Air leakage.

4.8.2.3.1 Suction. With one end of the coupling blanked, a suction of 15 in. WG. shall be applied to the other end. The overall leakage rate shall be not greater than that specified in 3.7.3.

4.8.2.3.2 Pressure. With one end of the coupling blanked, a pressure of 10 in. WG. shall be applied to the other end. The overall leakage rate shall be not greater than that specified in 3.7.3.

4.8.2.4 Pressure drop. An inlet pressure of 10 in. WG. shall be applied to the coupling. Flows shall be 25, 50 and 100 lpm. The pressure drop through the coupling shall be not greater than that specified in 3.7.4.

4.8.2.5 Water leakage. The self sealing valve shall be tested for water leakage at a hydrostatic head of 20 ft. WG. After a one minute exposure, the water leakage shall be measured and be as specified in 3.7.5.

4.8.2.6 Noise level. A ventilator which has passed the requirements of MIL-V-85830 or equal, shall be operated with a 28 volts d.c. power source connected to the external power jack. The noise shall be measured at the outlet hose. A coupling shall then be assembled to the outlet hose. The noise shall be measured at the outlet of the coupling. The noise shall be as specified in 3.7.6.

4.8.2.7 Chemical agent permeation. Rubber seals and o-rings from the coupling shall be tested for chemical agent permeation. Test slabs made of the same formulation, thickness, cure time and cure temperature as the rubber seals and o-rings shall be used in lieu of coupling materials.

4.8.2.7.1 Mustard. Resistance to permeation by mustard shall be tested as specified in MIL-STD-282, Method T-209. The material shall meet the requirements as specified in 3.7.7.

4.8.2.7.2 Vx. Resistance to permeation by Vx shall be tested as specified in MIL-STD-282, Method T-208. The material shall meet the requirements as specified in 3.7.7.

4.8.2.7.3 Thickened soman. Resistance to permeation by thickened soman shall be tested as specified in MIL-STD-282, Method T-208. The material shall meet the requirements as specified in 3.7.7.

4.8.3 Endurance. The coupling shall be cycled 500 times. A cycle shall consist of separation of the coupling at the quick disconnect fitting and reconnection at the quick disconnect fitting. At the conclusion of the 500 cycles, the coupling assembly shall meet the requirements of 3.7.8.

4.8.4 Environmental tests. Environmental tests shall be conducted in the sequence specified in Table VI. Prior to and at the completion of each environmental test, the coupling shall be tested as specified in 4.8.2.1, 4.8.2.2, 4.8.2.3, 4.8.2.4, and 4.8.2.5 and shall meet the requirements of 3.7.9.

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4.8.4.1 High temperature. The coupling shall be tested as specified in MIL-STD-810, Method 501.2, Procedure II, Operation. The coupling shall be placed in a climatic chamber. The chamber temperature shall be raised to 158°F at a rate of 5F° per minute. The chamber temperature shall be maintained at that temperature for a period of one hour. The coupling shall then be cycled one time (see 4.8.3). The relative humidity shall be not greater than 75 percent. The temperature shall then be reduced to 131°F at a rate of 5F° per minute and maintained at that temperature for a period, of six hours. At the completion of the six hour period, the coupling shall meet the requirements specified in 3.7.9 and 3.7.9.1.

4.8.4.2 Low temperature. The coupling shall be tested as specified in MIL-STD-810, Method 502.2, Procedure II, Operation. The coupling shall be placed in a climatic chamber. The chamber temperature shall be lowered to -40°F at a rate of 5F° per minute. The chamber temperature shall be maintained at that temperature for a period of one hour. The coupling shall then be cycled one time (see 4.8.3). The temperature shall then be brought to -14°F at a rate of 5F° per minute and maintained for a period of six hours. At the completion of the six hour period, the coupling shall meet the requirements specified in 3.7.9 and 3.7.9.2.

4.8.4.3 Salt fog. The coupling shall be tested as specified in MIL-STD-810, Method 509.2. The test shall consist of four alternating periods of 24 hours exposure and 24 hours drying time. The coupling shall meet the requirements specified in 3.7.9 and 3.7.9.3.

4.8.4.4 Icing and freezing rain. The coupling shall be tested as specified in MIL-STD-810, Method 521.0, Procedure I, Glaze Ice, Sea Spray condition. The coupling shall be cycled one time at 21°F (see 4.8.3). Ice may be broken to facilitate disconnection and connection. The coupling shall meet the requirements specified in 3.7.9 and 3.7.9.4.

4.8.4.5 Fungus. The coupling shall be tested as specified in MIL-STD-810, Method 508.3 for a period of 28 days. The coupling shall meet the requirements specified in 3.7.9 and 3.7.9.5.

4.8.4.6 Low pressure and high humidity. The coupling shall be tested as specified in MIL-STD-810, Method 500.2, Procedure II, Operation. The coupling shall be placed in a climatic chamber. The chamber temperature shall be reduced to -14°F at a rate not greater than 5°F per minute and the condition maintained for a period of 30 minutes. The pressure in the chamber shall then be reduced to .7 standard atmospheres over a period of 15 minutes and maintained at this pressure for 30 minutes. The chamber humidity shall then be raised and maintained at, or close to, saturation. The chamber temperature shall then be raised to 32°F and the pressure increased to NTP pressure at an approximately linear rate over a period of 15 minutes. The chamber temperature shall then be raised at an approximately linear rate to 86°F with the humidity maintained at or near saturation, over a period of 30 minutes. The chamber conditions shall be maintained for a period of not less than one hour and until all frost has melted. The coupling shall meet the requirements specified in 3.7.9 and 3.7.9.6.

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4.8.4.7 Dust. The coupling shall be tested as specified in MIL-STD-810, Method 510.2, Procedure I, Blowing Dust for a period of six hours at 73°F and 300 ft./min. air velocity and six hours at 131°F and 1750 ft./min. air velocity. Relative humidity shall be not greater than 30 percent. The coupling shall meet the requirements specified in 3.7.9 and 3.7.9.7.

4.8.4.8 Normal acceleration. The coupling shall be tested as specified in MIL-STD-810, Method 513.3, Procedure II, Operation. The acceleration shall be six g's in each of the six directions. The coupling shall meet the requirements specified in 3.7.9 and 3.7.9.8.

4.8.4.9 High temperature, diurnal cycle and high humidity. The coupling shall be tested as specified in MIL-STD-810, Method 507.2, Procedure II, Cycle 4, normal test duration. The coupling shall be examined every five days and shall pass the requirements specified in 3.7.9 and 3.7.9.9.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial (see 6.2.1).

5.1.1 Level A. The coupling assembly shall be preserved Method IC-3 in accordance with MIL-P-116.

5.1.1.1 Cleaning and drying. Cleaning shall be "C-1", any applicable process, in accordance with MIL-P-116 and any drying process of MIL-P-116 may be applied.

5.1.1.2 Unit packaging. The coupling assembly shall be cushioned with material conforming to PPP-C-795 or PPP-C-1842 and shall be placed in a sealed bag conforming to MIL-B-117, Types I, II or III, Classes A, B or C, Styles 1, 2 or 3. The bag shall then be placed into a PPP-B-566 unit container.

5.1.1.3 Intermediate packaging. When specified in the contract (see 6.2.1), units packaged in accordance with 5.1.1.2 shall be placed into an intermediate container conforming to PPP-B-636, Type CF, Class Weather Resistant Container. The number of unit containers per intermediate container and the number of intermediate containers per shipping container shall be specified in the contract (see 6.2.1). The intermediate container shall be closed as specified in PPP-B-636 specification.

5.1.2 Commercial. The coupling assembly shall be preserved and unit packed in accordance with ASTM D 3951.

5.2 Packing. Packing shall be Level A, B or Commercial as (see 6.2.1).

5.2.1 Level A. The coupling assembly, preserved in accordance with 5.1.1.2 and 5.1.1.3 shall be packed in containers conforming to Table I of MIL-STD-794, as specified for Level A.

5.2.2 Level B. The coupling assembly, preserved in accordance with 5.1.1.2 and 5.1.1.3 shall be packed in containers conforming to Table I of MIL-STD-794, as specified for Level B.

5.2.3 Commercial. The coupling assembly shall be, preserved and unit packed then packed as specified in ASTM D 3951.

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5.3 Marking. All unit, intermediate, and shipping containers shall be marked in accordance with MIL-STD-129, including bar code markings as specified in MIL-STD-129 and MIL-STD-1189.

6. NOTES

6.1 Intended use. This coupling forms a part of the PASS. The intended use for the coupling is to facilitate quick disconnection of the mask from the ventilator in emergency situations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Applicable Government part number.
c. Whether first article inspection is waived, or the name and address of the first article inspection laboratory and the name of the Government activity responsible for conducting the first article inspection program (see 4.4.1).
d. If quality conformance verification inspection is required, the name and address of the designated laboratory and the Goverment activity responsible for the verification program (see 4.6).
e. Applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements. Manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they might be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command (Code 51122), Department of the Navy, Washington, DC 20361-5110; however, authorization for qualification of products shall be obtained from the Commander, Naval Air Development Center, Warminster, PA 18974, (Code 6031). Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Naval Air Development Center (Code 6031) indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 3 complete couplings. The contracting officer should include specific instructions in acquisition documents, regarding arrangements for examinations, tests and approval of the first article. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.5 Key word listing.

Agent permeation
Agent resistant
Anti-drown
Biological
Chemical
Coupling
Decontamination
Quick disconnect
Radiological
Self sealing valve

Custodian:
Navy - AS

Preparing activity:
Navy - AS
(Project 4240-N880)

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Table I. Qualification inspection.

Inspection	Requirement	Test method	Sample number 1	2	3
Visual	-	4.8.1	X	X	X
Separation force/unlocked	3.7.1.1	4.8.2.1.1	X	X	X
Separation force/locked	3.7.1.2	4.8.2.1.2	X	X	X
Self sealing valve	3.7.2	4.8.2.2	X	X	X
Air leakage	3.7.3	4.8.2.3	X	X	X
Pressure drop	3.7.4	4.8.2.4	X	X	X
Water leakage	3.7.5	4.8.2.5	X	X	X
Noise level	3.7.6	4.8.2.6	X		
Agent permeation/mustard	3.7.7	4.8.2.7.1		-	
Agent permeation/Vx	3.7.7	4.8.2.7.2		-	
Agent permeation/thickened soman	3.7.7	4.8.2.7.3		-	
Endurance	3.7.8	4.8.3		X	
High temperature	3.7.9.1	4.8.4.1	See Table VI		
Low temperature	3.7.9.2	4.8.4.2	See Table VI		
Salt fog	3.7.9.3	4.8.4.3	See Table VI		
Icing and freezing rain	3.7.9.4	4.8.4.4	See Table VI		
Fungus	3.7.9.5	4.8.4.5	See Table VI		
Low pressure and high humidity	3.7.9.6	4.8.4.6	See Table VI		
Dust	3.7.9.7	4.8.4.7	See Table VI		
Normal acceleration	3.7.9.8	4.8.4.8	See Table VI		
High temperature, diurnal cycle and high humidity	3.7.9.9	4.8.4.9	See Table VI		

Table II. First article inspection.

Inspection	Requirement	Test method	Sample number 1	2	3
Visual	-	4.8.1	X	X	X
Separation force/unlocked	3.7.1.1	4.8.2.1.1	X	X	X
Separation force/locked	3.7.1.2	4.8.2.1.2	X	X	X
Self sealing valve	3.7.2	4.8.2.2	X	X	X
Air leakage	3.7.3	4.8.2.3	X	X	X
Pressure drop	3.7.4	4.8.2.4	X	X	X
Endurance	3.7.8	4.8.3	X	X	X

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Table III. Sample size, acceptance criteria, tests and examination of the coupling.

Inspection	Method	Sample size	Acceptance criteria
Air leakage	4.8.2.3	Normal level II S	AQL 0.65%
Pressure drop	4.8.2.4	Normal level II S	AQL 0.65%
Packaging	4.8.1.2	Normal level II S	AQL 2.5%
Visual	4.8.1	Normal level II S	AQL 2.5% for major AQL 4.0% for minor

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Table IV. Classification of defects for visual examination of the coupling assembly.

Major	Minor
101. Weight exceeds requirements.	201. Color not as specified.
102. Dimensions - overall length not as specified, diameter exceeds maximum.	202. Marking incorrect, missing, insufficient, illegible or not permanent.
103. Material imperfections - foreign matter embedded.	
104. Surfaces - unclean, rough, misaligned or containing cracks, nicks, or other flaws.	
105. Any component missing, malformed, fractured or otherwise damaged.	
106. Incorrect assembling or improper positioning of components.	
107. Any functioning part that works with difficulty.	
108. Faulty workmanship or other irregularity.	

Table V. List of defects for packaging.

Item	Defects
Exterior and interior markings.	Missing, incorrect, incomplete, illegible; of improper size, location, sequence; or method of application; markings not the same on the interior and exterior containers.
Packaging and packing materials.	Any non-conforming component; any component missing, damaged, or otherwise defective.
Workmanship.	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strapping, etc.; bulging or distortion of the containers.
Exterior and interior weight or content.	Number per container is more or less than required; gross or net weight exceeds the requirements.

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Table VI. Environmental qualification test sequence (coupling assembly).

Test	Unit Number						
	1	2	3	4	5	6	7
Visual	X	X	X	X	X	X	X
Low Pressure/High Humidity	X	X					
Low Temperature	X	X					
High Temperature	X	X					
Blowing Dust	X	X					
Fungus			X	X			
Salt Fog			X	X			
Acceleration					X	X	
High Temp/Diurnal/ High Humidity					X	X	
Icing and Freezing Rain							X

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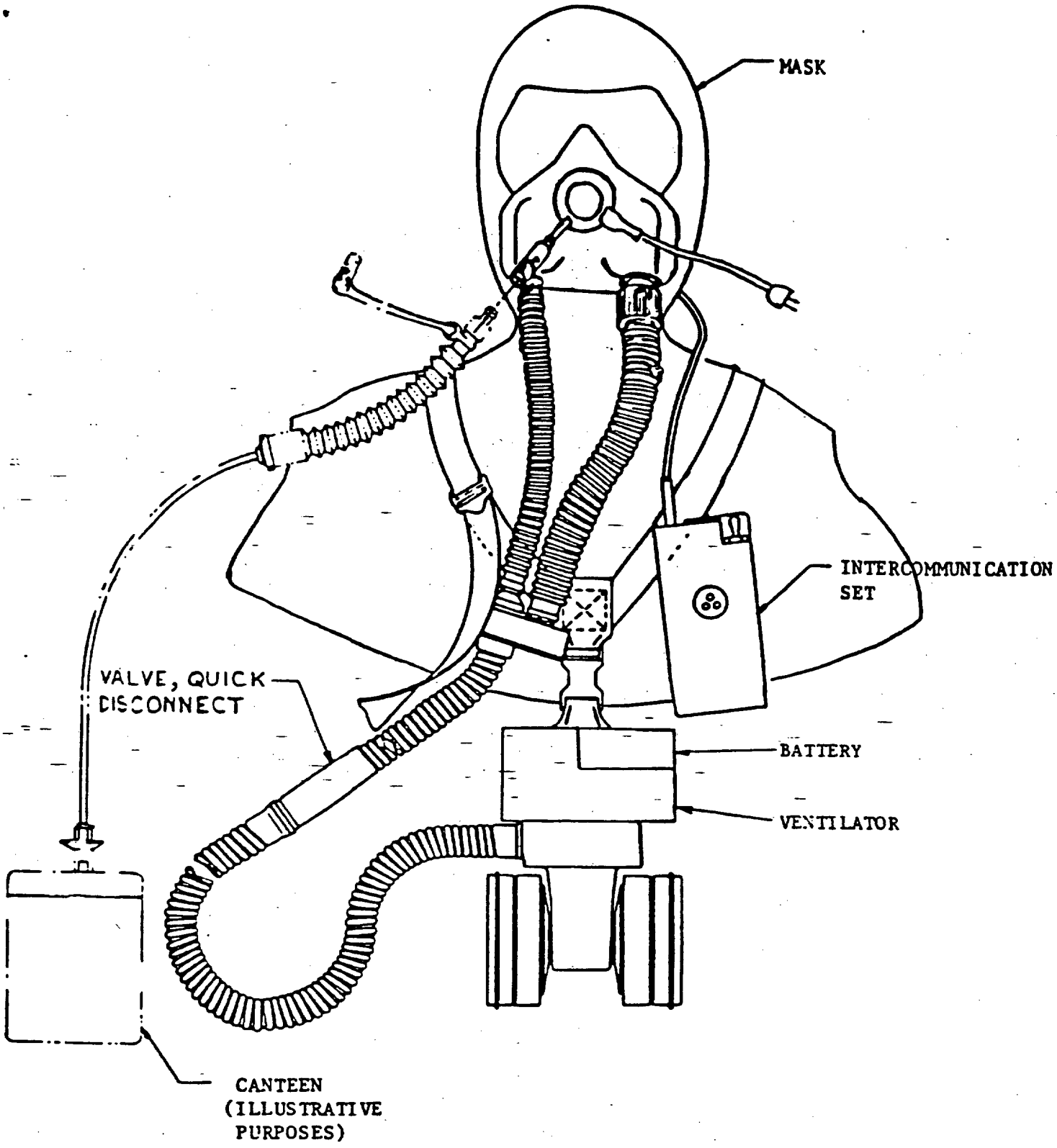


FIGURE 1. PROTECTIVE ASSEMBLY, CBR, HELICOPTER AIRCREWMEMBER

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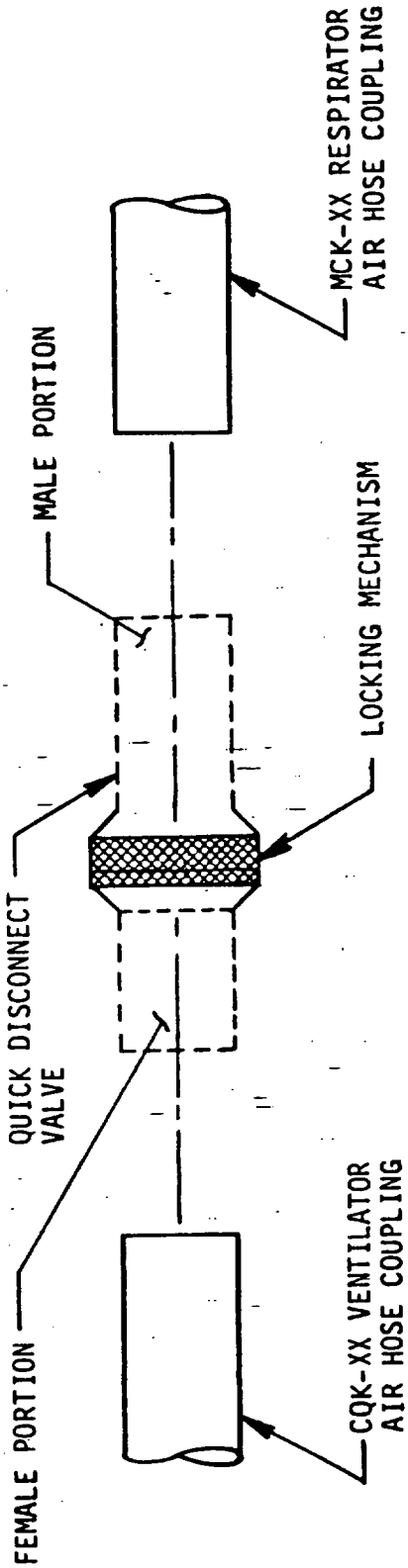


FIGURE 2. QUICK DISCONNECT VALVE

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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